QSAR and Graph Neural Network Modeling for Factor Xa Activity Prediction

Item	Value
Repository name	Factor-Xa-QSAR-GNN-Activity-Prediction
GitHub description	Scaffold-aware Factor Xa pKi prediction with ML baselines, graph neural networks, uncertainty estimation, and active-learning prioritization.
Target	Factor Xa / Coagulation Factor X
ChEMBL target ID	CHEMBL244
Modeled endpoint	pKi only
Final curated molecules	3,695
Primary evaluation	Scaffold split
Canonical model	Morgan_2048 + RandomForest

Executive Summary

This project builds a reproducible, scaffold-aware molecular machine learning workflow for predicting Factor Xa inhibitor potency from molecular structure. The workflow uses curated ChEMBL bioactivity data for Factor Xa / Coagulation Factor X and models pKi as the final regression target.

The project compares classical molecular ML baselines, RDKit descriptor models, combined feature models, and a PyTorch Geometric GINE graph neural network using fixed scaffold and random splits. The strongest scaffold-test model was a Morgan fingerprint Random Forest, which outperformed the graph neural network on this dataset.

After model selection, Random Forest tree-ensemble disagreement was used as a practical uncertainty proxy. This uncertainty signal correlated with absolute prediction error and modestly improved label efficiency in a retrospective active-learning simulation compared with random acquisition.

The project demonstrates applied molecular ML judgment: careful endpoint curation, leakage-aware scaffold evaluation, strong baseline benchmarking, honest GNN comparison, uncertainty analysis, and active-learning prioritization.

Scientific Goal and Key Questions

The goal was to predict Factor Xa inhibitor potency from molecular structure and evaluate whether model uncertainty could support active-learning prioritization.

1. Can curated public ChEMBL pKi data support a reliable Factor Xa potency prediction model?
2. Do graph neural networks outperform strong classical molecular ML baselines under scaffold-aware evaluation?
3. Can Random Forest ensemble disagreement identify molecules with higher prediction error?
4. Can uncertainty-based acquisition improve label efficiency compared with random acquisition?

Dataset and Curation

The dataset was built from public ChEMBL bioactivity data for Factor Xa / Coagulation Factor X using target ID CHEMBL244.

Only pKi values were modeled. Ki, IC50, and other endpoint types were not mixed into the final regression target, reducing endpoint heterogeneity and improving interpretability.

Molecules were curated at the structure level, represented as standardized SMILES, and split into fixed train, validation, and test partitions before model training.

Split	Molecules	Purpose
Scaffold train	2,586	Model fitting and active-learning acquisition pool
Scaffold validation	554	Model selection and hyperparameter decisions
Scaffold test	555	Final headline evaluation on held-out scaffolds

Workflow Overview

Step	Script	Purpose
01	01_pull_fxa_chembl_raw.py	Download raw Factor Xa ChEMBL bioactivity data
02	02_compare_fxa_endpoints.py	Compare endpoint availability and select final endpoint strategy
03	03_curate_fxa_dataset.py	Curate the final pKi regression dataset
04	04_make_features_and_splits.py	Generate Morgan fingerprints and fixed scaffold/random splits
05	05_train_baselines_morgan.py	Train Morgan fingerprint RF/XGBoost/LightGBM baselines
05b	05b_benchmark_morgan_rdkit2d_combined_features.py	Benchmark Morgan, RDKit 2D, and combined feature sets
06	06_train_gnn_fxa.py	Train the PyTorch Geometric GINE graph neural network
07	07_uncertainty_error_analysis.py	Analyze RF uncertainty versus prediction error
08	08_active_learning_uncertainty_vs_random.py	Run retrospective active-learning simulation

Central design rule: all model comparisons used the same saved splits. This avoids split drift and makes the model comparisons defensible.

Feature Sets and Models

Category	Methods
Molecular representations	Morgan_2048 fingerprints; RDKit_2D_raw descriptors; Morgan_plus_RDKit_2D_raw combined features
Classical ML models	Random Forest, XGBoost, LightGBM
Graph neural network	PyTorch Geometric GINE model with atom and bond categorical embeddings
Preprocessing control	Descriptor imputation and zero-variance filtering were fit only on the training split using scikit-learn pipelines

Key Results

1. Canonical scaffold-test model

Feature set	Model	RMSE	MAE	R2	Spearman
Morgan_2048	RandomForest	0.973	0.728	0.661	0.813

Morgan_2048 + RandomForest was selected as the canonical downstream model because it provided the strongest scaffold-test generalization among the tested feature/model combinations.

2. Feature benchmark interpretation

Feature/model observation	Interpretation
Morgan fingerprints outperformed RDKit 2D descriptors	Circular fingerprints captured the most useful structure-activity signal for this curated dataset.
Morgan + RDKit 2D did not improve scaffold generalization	Adding descriptors increased feature complexity without improving held-out scaffold performance.
Random split performance exceeded scaffold split performance	Random splits likely overestimate generalization when close analogs are shared between train and test sets.

GNN Benchmark, Uncertainty, and Active Learning

3. GNN benchmark

Model	Representation	RMSE	MAE	R2	Spearman
RandomForest	Morgan_2048	0.973	0.728	0.661	0.813
GINE GNN	Molecular graph	1.056	0.790	0.601	0.775

The GINE graph neural network learned meaningful structure-activity signal, but it did not outperform the Morgan Random Forest baseline. This is a useful and honest result: for this dataset, a strong classical molecular fingerprint baseline generalized better to held-out scaffolds than the tested GNN.

4. Uncertainty analysis

Metric	Value
Spearman correlation between RF uncertainty and absolute error	0.275
p-value	4.1e-11
Top 5% uncertainty enrichment for high-error molecules	2.7x

Random Forest tree-level prediction standard deviation was used as an ensemble-disagreement score, not as a calibrated Bayesian uncertainty estimate. The positive correlation with absolute error supports its use as a practical flag for molecules requiring expert review or additional labeling.

5. Active-learning simulation

Acquisition strategy	Final labeled molecules	Test RMSE	Test MAE	Test Spearman
Random	1,259	1.028	0.775	0.790
Uncertainty	1,259	0.996	0.753	0.805

At the same labeling budget, uncertainty sampling reduced final RMSE by 0.032 pKi, reduced MAE by 0.022 pKi, and improved Spearman correlation by 0.0155. The improvement was modest but meaningful because it connects uncertainty estimation to a practical compound acquisition strategy.

Main Findings

1. Morgan fingerprints were the strongest molecular representation among the tested feature sets.
2. Morgan_2048 + RandomForest was the best scaffold-selected model, with scaffold-test RMSE = 0.973 and Spearman = 0.813.

3. Random split performance was higher than scaffold split performance, confirming that random splits can overestimate generalization in molecular datasets.
4. The GINE graph neural network learned useful signal but did not outperform the classical Morgan Random Forest baseline.
5. Random Forest ensemble disagreement provided useful uncertainty information and improved label efficiency in a retrospective active-learning simulation.

Recommended Figures for GitHub or Manuscript Use

Figure file	Purpose
fxa_05b_scaffold_test_rmse_by_feature_set.png	Feature/model comparison by scaffold-test RMSE
fxa_05b_best_scaffold_test_pred_vs_actual.png	Canonical model predicted vs actual scaffold-test plot
fxa_06_scaffold_test_pred_vs_actual.png	GNN scaffold-test predicted vs actual plot
fxa_07_scaffold_test_abs_error_vs_uncertainty.png	Relationship between RF uncertainty and absolute prediction error
fxa_07_uncertainty_deciles_abs_error.png	Prediction error by uncertainty bin
fxa_08_al_test_rmse_curve.png	Active-learning RMSE curve; strongest final figure for the workflow story
fxa_08_al_test_spearman_curve.png	Active-learning Spearman curve

Reproducibility

Run the workflow from the project root in the following order:

```
python .\scripts\01_pull_fxa_chembl_raw.py *> .\scripts\01.log
python .\scripts\02_compare_fxa_endpoints.py *> .\scripts\02.log
python .\scripts\03_curate_fxa_dataset.py *> .\scripts\03.log
python .\scripts\04_make_features_and_splits.py *> .\scripts\04.log
python .\scripts\05_train_baselines_morgan.py *> .\scripts\05.log
python .\scripts\05b_benchmark_morgan_rdkit2d_combined_features.py *>
.\scripts\05b.log
python .\scripts\06_train_gnn_fxa.py *> .\scripts\06.log
python .\scripts\07_uncertainty_error_analysis.py *> .\scripts\07.log
python .\scripts\08_active_learning_uncertainty_vs_random.py *> .\scripts\08.log
```

Output folder	Contents
results/metrics/	JSON summaries and run-level metrics
results/tables/	Curated datasets, model summaries, active-learning tables
results/figures/	Prediction plots, benchmark plots, uncertainty plots, active-learning curves
models/	Saved model artifacts, if included or released separately

Limitations

1. Public ChEMBL assay data can contain assay heterogeneity even after endpoint filtering.
2. Only pKi values were modeled to reduce endpoint mixing; experimental conditions may still vary across assays.
3. Scaffold splitting is stricter than random splitting but remains a retrospective approximation of prospective medicinal chemistry deployment.
4. Random Forest tree standard deviation is an uncertainty proxy, not a fully calibrated uncertainty model.

5. The active-learning experiment is retrospective and should ideally be validated prospectively with newly acquired experimental data.
6. The GNN was used as a benchmark, but extensive architecture tuning was not the main focus because the tested model did not outperform the classical baseline.

Final Conclusion

This project demonstrates a complete scaffold-aware molecular ML workflow for Factor Xa potency prediction. A Morgan fingerprint Random Forest provided the strongest scaffold-test performance among the tested feature/model combinations, outperforming RDKit descriptor models, combined feature models, and the tested PyTorch Geometric GINE graph neural network.

Random Forest ensemble disagreement was a useful practical uncertainty signal and produced modest label-efficiency gains in a fixed-test active-learning simulation. The key scientific lesson is that strong classical baselines remain essential in molecular machine learning, even when graph neural networks are included.

Overall, the project is suitable as a professional GitHub portfolio project because it demonstrates data curation, endpoint discipline, leakage-aware evaluation, scaffold-based generalization, GNN benchmarking, uncertainty analysis, and active-learning decision support.